Understanding the Performance of AI Algorithms in Text-Based Emotion Detection for Conversational Agents

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Project Guide
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Introduction

Text-based emotion detection is the process of identifying and interpreting human emotions from textual conversations, enabling machines to understand users' emotional states. It is needed to make conversational agents more empathetic, responsive, and human-like, improving user satisfaction and engagement. Applications of text-based emotion detection include chatbots, virtual assistants, customer support systems, e-learning platforms, healthcare advisory systems, and social media analysis, where understanding user emotions enhances interaction quality and decision-making.

Literature Survey

Author(s)	Method	Advantages	Disadvantages
Y. Shou, T. Meng, W. Ai, S. Yang, and K. Li	Conversational emotion recognition studies	High accuracy using multiple modalities	Complex preprocessing
J. Devlin et al. [2]	BERT	Captures context; adaptable	Large model; high computation
G. Tu, J. Wen, C. Liu, D. Jiang, and E. Cambria	Context- and sentiment-aware networks	Good generalization; identifies strengths	Varies across datasets
N. Nguyen, T.-H. Nguyen, Y.-N. Nguyen, D. Doan, M. Nguyen, and V.-H. Nguyen	Machine learning-based model	Generates empathetic responses	Limited on rare emotions

Problem Statement

Text-based emotion detection enables conversational agents to recognize and respond to users' emotional states effectively. Existing studies employ machine learning, deep learning, and pre-trained language models for emotion classification, yet their performance varies across datasets and emotional categories. Most research lacks a comprehensive comparison of these approaches, making model selection challenging. To address this issue, this study systematically evaluates multiple AI algorithms using benchmark datasets to analyze their performance and robustness, contributing to the development of empathetic conversational agents.

Objectives

1. **To analyze user text and identify underlying emotions** using machine learning and natural language processing techniques.

2. **To evaluate and compare different Al algorithms** for accuracy and performance in emotion detection.

3. **To improve conversational agents** by enabling emotion-aware responses for more natural and effective human-computer interaction.

Methodology

The proposed methodology involves a comparative analysis of multiple AI algorithms for text-based emotion detection in conversational agents. The process begins with data preprocessing, including text cleaning, tokenization, and feature representation. Traditional machine learning models, deep learning architectures, and pre-trained language models are then trained and evaluated on benchmark datasets. Model performance is assessed using standard evaluation metrics such as accuracy, precision, recall, and F1-score. The comparative results are analyzed to identify the most effective algorithms for reliable and emotionally aware conversational systems.

Project Time Line

Functionality	Status
Literature Survey and Problem Definition	Completed
Dataset Preparation and Preprocessing	Completed
Model Development and Implementation	In Progress
Model Training and Performance Evaluation	In Progress
Result Analysis and Comparison	Not Started
Report Documentation and Submission	Not Started

References

1. J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectionaltransformers for language understanding,” in Proc. NAACL-HLT, 2019, pp. 4171–4186.
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4. N. Nguyen, T.-H. Nguyen, Y.-N. Nguyen, D. Doan, M. Nguyen, and V.-H. Nguyen. 2023. Machine learning-based model for customer emotion detection in hotel booking services. Journal of Hospitality and Tourism Insights. Published July 17, 2023.

THANK YOU